

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2025-2026

Program: B.Tech Stream: Mechatronics

Year: 2nd Semester: IV

Subject: Dynamic Systems Modeling (702MH0C023)

Time: 45 Minutes

Date:

No. of Pages: 1

Marks: 10

Set: 1

Re-Examination

Instructions: Candidates should read carefully the instructions.

- 1) Figures in brackets on the right hand side indicate full marks.
- 2) Syllabus scope: Unit 1 – Unit 5 (complete).
- 3) Assume suitable data if necessary.
- 4) Question 1 is compulsory.
- 5) Answer any 2 from the remaining questions.

Q1

Answer briefly:

CO- 1;
SO- 1;
BL- 1,2

- a. (i) Write the expressions for velocity in polar coordinates (v_r and v_θ components) and state [2]
when polar coordinates are preferred over Cartesian.
(ii) Define the Lagrangian L of a mechanical system and state the Euler-Lagrange equation in terms of a generalized coordinate q .
(iii) State the transport theorem for a vector quantity as observed from a fixed (inertial) frame and a rotating frame. Identify the correction term.
(iv) Define the damping ratio ζ and state the three response regimes (underdamped, critically damped, overdamped) with their conditions.
- b. (i) A particle moves in a circle of radius $r = 2$ m at constant angular speed $\dot{\theta} = 3$ rad/s. Calculate [2]
the radial (centripetal) acceleration a_r and the transverse acceleration a_θ .
(ii) Write the kinetic energy T and potential energy V for a simple pendulum of mass m and length L , using θ as the generalized coordinate.
(iii) Define a holonomic constraint and state how the number of degrees of freedom (DOF) is determined for a system of N particles with k holonomic constraints.
(iv) A spring-mass system has $m = 4$ kg and $k = 256$ N/m. Calculate the natural frequency ω_n (rad/s) and the time period T of free oscillation.

Q2

CO- 1,2;
SO- 1,2;
BL- 3

A simple pendulum consists of a point mass $m = 1$ kg attached to a massless, inextensible string of [3]
length $L = 1$ m. The bob swings in a vertical plane under gravity ($g = 9.81$ m/s²). Using the angle θ
from the vertical as the generalized coordinate, analyze the system using the Lagrangian method.
Use the following relations: Lagrangian $L = T - V$; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin \theta \approx \theta$.

(a) State the number of degrees of freedom of this system and identify the generalized coordinate.
(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L .
(c) Derive the kinetic energy T in terms of $\theta, \dot{\theta}, m$, and L .
(d) Write the potential energy V in terms of θ, m, g , and L .
(e) Form the Lagrangian $L = T - V$ and apply the Euler-Lagrange equation to derive the equation of motion.
(f) For small oscillations ($\sin \theta \approx \theta$), determine the natural frequency of the pendulum.

Q3

CO- 2;
SO- 2;
BL- 3,4

A horizontal platform rotates at a constant angular velocity $\Omega = 4$ rad/s about a vertical axis. An [3]
object at a radial distance $r = 1.5$ m from the center moves radially outward at a constant speed
 $v_{rel} = 2$ m/s ($\dot{r} = 2$ m/s, $\ddot{r} = 0$). The angular acceleration of the platform is zero ($\alpha = 0$).
(a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term.
(b) Calculate the centripetal acceleration magnitude at $r = 1.5$ m.
(c) Calculate the Coriolis acceleration magnitude.
(d) State the direction of the Coriolis acceleration relative to the radial velocity.
(e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame.
(f) If the object continues at the same radial speed, find the centripetal acceleration when $r = 3$ m and comment on how it changes.

Q4

CO- 2,3;
SO- 2,3;
BL- 4,5

A mass $m = 3 \text{ kg}$ is attached to a spring of stiffness $k = 300 \text{ N/m}$ and a viscous damper with damping coefficient $c = 12 \text{ N}\cdot\text{s/m}$. The system undergoes free vibration after being displaced 0.05 m from equilibrium and released from rest. Take the equilibrium position as the origin. [3]

- (a) Write the equation of motion for this damped spring-mass system.
- (b) Calculate the natural frequency ω_n and the critical damping coefficient c_{cr} .
- (c) Calculate the damping ratio ζ and state the type of damping.
- (d) Calculate the damped natural frequency ω_d .
- (e) Calculate the logarithmic decrement δ and find the ratio of two successive peak amplitudes.
- (f) If the damper is removed ($c = 0$), write the displacement as a function of time $x(t)$ for the given initial conditions.

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1

Answer the following:

- (a) Write the expressions for velocity in polar coordinates (v_r and v_θ components) and state when polar coordinates are preferred over Cartesian. [0.5]

Ans:

In polar coordinates:

$$v_r = \dot{r} \text{ (radial component)}$$

$$v_\theta = r\dot{\theta} \text{ (transverse component)}$$

$$\rightarrow \mathbf{v} = \dot{r} \hat{e}_r + r\dot{\theta} \hat{e}_\theta$$

Polar coordinates are preferred when the motion has radial symmetry or when dealing with central-force problems (e.g. planetary motion, pendulums), because the constraint is naturally expressed in r and θ .

- (b) Define the Lagrangian L of a mechanical system and state the Euler-Lagrange equation in terms of a generalized coordinate q . [0.5]

Ans:

The Lagrangian is defined as $L = T - V$, where T = kinetic energy and V = potential energy.

The Euler-Lagrange equation is:

$$d/dt (\partial L / \partial \dot{q}) - \partial L / \partial q = 0$$

This yields the equation(s) of motion for the system in terms of the generalized coordinate q .

- (c) State the transport theorem for a vector quantity as observed from a fixed (inertial) frame and a rotating frame. Identify the correction term. [0.5]

Ans:

The transport theorem states:

$$(d\mathbf{A}/dt)_{\text{fixed}} = (d\mathbf{A}/dt)_{\text{rotating}} + \boldsymbol{\Omega} \times \mathbf{A}$$

where $\boldsymbol{\Omega}$ is the angular velocity of the rotating frame.

The correction term $\boldsymbol{\Omega} \times \mathbf{A}$ accounts for the rotation of the basis vectors themselves as seen from the fixed frame.

- (d) Define the damping ratio ζ and state the three response regimes (underdamped, critically damped, overdamped) with their conditions. [0.5]

Ans:

The damping ratio $\zeta = c / c_{cr} = c / (2\sqrt{km})$, where c = damping coefficient, k = stiffness, m = mass.

Three regimes:

$\zeta < 1 \rightarrow$ Underdamped: oscillatory with exponentially decaying amplitude.

$\zeta = 1 \rightarrow$ Critically damped: fastest return to equilibrium without overshoot.

$\zeta > 1 \rightarrow$ Overdamped: slow, non-oscillatory return to equilibrium.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A particle moves in a circle of radius $r = 2$ m at constant angular speed $\dot{\theta} = 3$ rad/s. Calculate the radial (centripetal) acceleration a_r and the transverse acceleration a_θ . [0.5]

Ans:

For circular motion at constant angular speed ($\dot{r} = 0$, $\ddot{r} = 0$, $\dot{\theta} = 3$):

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 2 \times 3^2 = -18 \text{ m/s}^2 \text{ (directed radially inward)}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 0 = 0 \text{ m/s}^2$$

The magnitude of centripetal acceleration is 18 m/s^2 .

- (b) Write the kinetic energy T and potential energy V for a simple pendulum of mass m and length L , using θ as the generalized coordinate. [0.5]

Ans:

For a simple pendulum with angle θ :

$$T = \frac{1}{2} m (L\dot{\theta})^2 = \frac{1}{2} m L^2 \dot{\theta}^2$$

$$V = -mgL \cos \theta \text{ (taking the pivot as the datum)}$$

or equivalently $V = mgL(1 - \cos \theta)$ with datum at the lowest point.

The Lagrangian is $L = T - V = \frac{1}{2}mL^2\dot{\theta}^2 + mgL \cos \theta$.

(c) Define a holonomic constraint and state how the number of degrees of freedom (DOF) is determined for a system of N particles with k holonomic constraints. [0.5]

Ans:

A holonomic constraint is one that can be expressed as an equation relating only coordinates (and possibly time), not velocities: $f(q_1, q_2, \dots, t) = 0$.

DOF = $3N - k$, where N = number of particles, k = number of independent holonomic constraints.

Example: A bead on a wire ($k = 2$ of 3 coordinates constrained $\rightarrow$ 1 DOF).

(d) A spring-mass system has $m = 4$ kg and $k = 256$ N/m. Calculate the natural frequency ω_n (rad/s) and the time period T of free oscillation. [0.5]

Ans:

$$\omega_n = \sqrt{k/m} = \sqrt{256/4} = \sqrt{64} = 8 \text{ rad/s.}$$

$$T = 2\pi/\omega_n = 2\pi/8 = \pi/4 \approx 0.785 \text{ s.}$$

Solve any TWO from Q2, Q3, Q4 [3 Marks each = 6 Marks]

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO1, CO2

A simple pendulum consists of a point mass $m = 1$ kg attached to a massless, inextensible string of length $L = 1$ m. The bob swings in a vertical plane under gravity ($g = 9.81$ m/s²). Using the angle θ from the vertical as the generalized coordinate, analyze the system using the Lagrangian method. Use the following relations: Lagrangian $L = T - V$; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin \theta \approx \theta$.

(a) State the number of degrees of freedom of this system and identify the generalized coordinate. [0.5]

Ans:

The pendulum has 1 degree of freedom (DOF = 1).

Generalized coordinate: θ — the angle of the string measured from the vertical.

(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L . [0.5]

Ans:

Taking the pivot as the origin, with x horizontal and y vertically downward:

$$x = L \sin \theta = 1.0 \sin \theta$$

$$y = L \cos \theta = 1.0 \cos \theta$$

$$\text{Differentiating: } \dot{x} = L\dot{\theta} \cos \theta, \dot{y} = -L\dot{\theta} \sin \theta$$

(c) Derive the kinetic energy T in terms of $\theta, \dot{\theta}, m$, and L . [0.5]

Ans:

$$v^2 = \dot{x}^2 + \dot{y}^2 = L^2\dot{\theta}^2 \cos^2 \theta + L^2\dot{\theta}^2 \sin^2 \theta = L^2\dot{\theta}^2$$

$$T = \frac{1}{2}m\dot{y}^2 = \frac{1}{2} \times 1.0 \times 1.0^2 \times \dot{\theta}^2 = \frac{1}{2}mL^2\dot{\theta}^2$$

$$T = \frac{1}{2}(1.0)(1.0^2)\dot{\theta}^2 = 0.5 \dot{\theta}^2 \text{ J}$$

(d) Write the potential energy V in terms of θ, m, g , and L . [0.5]

Ans:

Taking the pivot as the datum (y positive downward):

$$V = -mgy = -mgL \cos \theta = -1.0 \times 9.81 \times 1.0 \times \cos \theta$$

$$V = -9.81 \cos \theta \text{ J}$$

(e) Form the Lagrangian $L = T - V$ and apply the Euler-Lagrange equation to derive the equation of motion. [0.5]

Ans:

$$L = T - V = \frac{1}{2}mL^2\dot{\theta}^2 + mgL \cos \theta$$

$$\text{Euler-Lagrange: } d/dt(\partial L/\partial \dot{\theta}) - \partial L/\partial \theta = 0$$

$$\partial L/\partial \dot{\theta} = mL^2\dot{\theta}$$

$$d/dt(mL^2\dot{\theta}) = mL^2\ddot{\theta}$$

$$\partial L/\partial \theta = -mgL \sin \theta$$

$$\text{EOM: } mL^2\ddot{\theta} + mgL \sin \theta = 0$$

$$\text{Simplifying: } \ddot{\theta} + (g/L) \sin \theta = 0$$

(f) For small oscillations ($\sin \theta \approx \theta$), determine the natural frequency of the pendulum. [0.5]

Ans:

$$\text{For small angles, } \sin \theta \approx \theta \rightarrow \ddot{\theta} + (g/L)\theta = 0$$

This is SHM with $\omega_n^2 = g/L$

$$\omega_n = \sqrt{g/L} = \sqrt{9.81/1.0} = \sqrt{9.81} = 3.132 \text{ rad/s}$$

$$\text{Time period } T = 2\pi/\omega_n = 2.006 \text{ s} \approx 2.01 \text{ s}$$

Q3 [3 Marks]

Bloom: BL 3–4 (Apply / Analyze) | CO2

A horizontal platform rotates at a constant angular velocity $\Omega = 4 \text{ rad/s}$ about a vertical axis. An object at a radial distance $r = 1.5 \text{ m}$ from the center moves radially outward at a constant speed $v_{\text{rel}} = 2 \text{ m/s}$ ($\dot{r} = 2 \text{ m/s}$, $\ddot{r} = 0$). The angular acceleration of the platform is zero ($\alpha = 0$).

- (a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term. [0.5]

Ans:

The fixed-frame (inertial) acceleration, expressed in terms of rotating-frame quantities, is:

$$\mathbf{a}_{\text{fixed}} = \mathbf{a}_{\text{rel}} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$$

where:

$\mathbf{a}_{\text{rel}}$ = acceleration of the object in the rotating frame ($\ddot{r} = 0$ here),

$2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ = Coriolis acceleration,

$\boldsymbol{\alpha} \times \mathbf{r}$ = Euler (angular acceleration) term (zero here since $\alpha = 0$),

$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ = centripetal acceleration (directed radially inward).

- (b) Calculate the centripetal acceleration magnitude at $r = 1.5 \text{ m}$. [0.5]

Ans:

$$a_{\text{centripetal}} = \Omega^2 r = 4.0^2 \times 1.5 = 16 \times 1.5 = 24.0 \text{ m/s}^2$$

Direction: radially inward (toward the axis of rotation).

- (c) Calculate the Coriolis acceleration magnitude. [0.5]

Ans:

$$a_{\text{Coriolis}} = 2\Omega v_{\text{rel}} = 2 \times 4.0 \times 2.0 = 16.0 \text{ m/s}^2$$

- (d) State the direction of the Coriolis acceleration relative to the radial velocity. [0.5]

Ans:

By the right-hand rule for $2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ ($\boldsymbol{\Omega}$ along vertical, $\mathbf{v}_{\text{rel}}$ radially outward): the Coriolis acceleration is perpendicular to $\mathbf{v}_{\text{rel}}$, directed tangentially in the direction opposite to the platform's rotation. This is why a ball thrown outward on a rotating turntable curves sideways.

- (e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame. [0.5]

Ans:

$$\begin{aligned} |a_{\text{total}}| &= \sqrt{a_{\text{cent}}^2 + a_{\text{Cor}}^2} = \sqrt{(24.0)^2 + (16.0)^2} \\ &= \sqrt{576 + 256} = \sqrt{832} = 28.84 \text{ m/s}^2 \end{aligned}$$

- (f) If the object continues at the same radial speed, find the centripetal acceleration when $r = 3 \text{ m}$ and comment on how it changes. [0.5]

Ans:

$$\text{At } r = 3 \text{ m: } a_{\text{centripetal}} = \Omega^2 r = 4.0^2 \times 3 = 48.0 \text{ m/s}^2$$

Compared to 24.0 m/s^2 at $r = 1.5 \text{ m}$, the centripetal acceleration has doubled. It increases linearly with radial distance ($a_{\text{cent}} \propto r$), while the Coriolis acceleration (16.0 m/s^2) remains the same since v_{rel} is unchanged.

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A mass $m = 3 \text{ kg}$ is attached to a spring of stiffness $k = 300 \text{ N/m}$ and a viscous damper with damping coefficient $c = 12 \text{ N}\cdot\text{s/m}$. The system undergoes free vibration after being displaced 0.05 m from equilibrium and released from rest. Take the equilibrium position as the origin.

- (a) Write the equation of motion for this damped spring-mass system. [0.5]

Ans:

The equation of motion for a damped SDOF system:

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$3.0\ddot{x} + 12.0\dot{x} + 300.0x = 0$$

$$\text{Dividing by } m: \ddot{x} + 4.0\dot{x} + 100.0x = 0$$

$$\text{Or: } \ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$$

- (b) Calculate the natural frequency ω_n and the critical damping coefficient c_{cr} . [0.5]

Ans:

$$\omega_n = \sqrt{k/m} = \sqrt{300.0/3.0} = \sqrt{100} = 10.00 \text{ rad/s}$$

$$c_{\text{cr}} = 2\sqrt{km} = 2\sqrt{300.0 \times 3.0} = 2\sqrt{900} = 2 \times 30.00 = 60.00 \text{ N}\cdot\text{s/m}$$

- (c) Calculate the damping ratio ζ and state the type of damping. [0.5]

Ans:

$$\zeta = c / c_{\text{cr}} = 12.0 / 60.00 = 0.2000$$

Since $\zeta = 0.2000 < 1 \rightarrow$ the system is UNDERDAMPED.

It will oscillate with exponentially decaying amplitude.

(d) Calculate the damped natural frequency ω_d . [0.5]

Ans:

$$\begin{aligned}\omega_d &= \omega_n \sqrt{1 - \zeta^2} = 10.00 \times \sqrt{1 - 0.2000^2} \\ &= 10.00 \times \sqrt{1 - 0.040000} \\ &= 10.00 \times 0.979796 \\ &= 9.7980 \text{ rad/s} \approx 9.80 \text{ rad/s}\end{aligned}$$

(e) Calculate the logarithmic decrement δ and find the ratio of two successive peak amplitudes. [0.5]

Ans:

$$\begin{aligned}\delta &= 2\pi\zeta / \sqrt{1 - \zeta^2} = 2\pi \times 0.2000 / \sqrt{1 - 0.040000} \\ &= 1.2566 / 0.979796 \\ &= 1.2825\end{aligned}$$

$$\text{Ratio of successive peaks: } x_n / x_{(n+1)} = e^\delta = e^{1.2825} = 3.6058$$

$$\text{Or equivalently } x_{(n+1)} / x_n = e^{-\delta} = 0.2773$$

Each successive peak is $\approx 27.73\%$ of the previous one.

(f) If the damper is removed ($c = 0$), write the displacement as a function of time $x(t)$ for the given initial conditions. [0.5]

Ans:

With $c = 0$ (undamped), EOM: $m\ddot{x} + kx = 0$

$$x(t) = x_0 \cos(\omega_n t) = 0.05 \cos(10.00 t) \text{ m}$$

(Since initial conditions are $x(0) = 0.05 \text{ m}$, $\dot{x}(0) = 0$, the solution is a pure cosine with amplitude 0.05 m and period $T = 2\pi/10.00 = 0.6283 \text{ s}$)

— End of Model Answers —